

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (C.E./ I.T.)

May-2014

CE403 Data Warehousing & Data Mining (DW&DM)

Date: 22.05.2014, Thursday Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Rough work is to be done in the last page of main supplementary, please don't write anything on the question paper.
5. Indicate clearly, the option(s) you attempt along with its respective question no.

SECTION-I

Q-1 Answer the following questions.

- a. Briefly describe each types of Concept Hierarchy with an example. 3
- b. What do you mean by normalization in data warehousing and mining? How is it differ from the concept of database? 3
- c. Give architecture of data warehouse. List out the components of data warehouse. 3
- d. How to find the task relevant data from the domain of interest? 2

Q-2

[A] Association Rule Mining often generates a large number of rules. Discuss-effective methods that can be used to reduce the number of rules generated while still preserving most of the interesting rules. 4

[B] Compare and contrast between the incremental update operation performed in ROLAP, MOLAP and HOLAP. 4

OR

[B] Explain the role of star network query model in querying multidimensional databases with an example. 4

[C] Suppose that the data for analysis include the attribute age. The age values for the data tuples are: 4

13,15,16,16,19,20,20,21,22,22,25,25,25,25,25,30,33,35,35,35,35,36,40,45,46,52,70.

- a. Use min-max normalization to transform the value 35 for age on to 0-1 range
- b. Use z-score normalization to transform value 35, where std. Deviation is 12.94 yrs.
- c. Use normalization by decimal scaling to transform value 35 for age.

OR

[C] What is an iceberg query? Can we use it in Market Basket Analysis? Justify your answer with an example. 4

Q-3

[A] Is Data Warehouse a prerequisite for Data Mining? In data mining task quality of data is more important than the quantity of the data. Justify. 4

OR

[A] Suppose that a data warehouse consists of the three dimensions time, doctor and patient, and two measures count and charge 4

1. Draw a schema diagram
2. Give OLAP operations in order to list the total fee collected by each doctor in 2000?

[B] Why fact constellation schema model is not suitable for data marts? Which process model is suitable for designing data marts? 4

OR

[B] Where are supervised and unsupervised learning used and what purpose do they serve? Is clustering a supervised or an unsupervised form of learning? What do you mean by shape of the cluster? What is the most common shape? 4

[C] Compare the advantages and disadvantages of eager classification (e.g., decision tree, Bayesian, neural network) versus lazy classification (e.g., k-nearest neighbor, case-based reasoning). 4

OR

[C] Why star schema model is more useful for join indexing? How to calculate a measure value at a given point? 4

SECTION-II

Q-4

1. Differentiate between: Text Mining and Web Mining. 3
2. Explain the role of Fuzzy Set Theory and Genetic Algorithms in practical Data Mining. 4
3. When a decision tree is built, many of the branches will reflect anomalies in the training data due to noise or outliers and will decrease performance of the classifier. Explain the approaches to address above problem. 4

Q-5

- [A] What are multidimensional associational rules? Explain in brief. 4
- [B] Explain in brief about Attribute Oriented Induction (AOI) technique. Compare it with data cube based implementation approach. 4
- [C] Differentiate between gini index and entropy based evaluation technique for relevance of the attributes? 4

OR

Q-5

- [A] Why is tree pruning useful in Decision Tree Induction? What is a drawback of using a separate set of tuples to evaluate pruning? 4
- [B] Write an algorithm for k-nearest neighbor classification given k and n, the number of attributes describing each sample. 4
- [C] Following table consists of training data from an employee database. The data have been generalized. For a given row entry, count represents the number of data tuples having the values for department, status, age, and salary given in that row. Let salary be the class label attribute. 4

department	status	age	salary	count
sales	senior	31-35	45-50K	30
sales	junior	26-30	25-30K	40
sales	junior	31-35	30-35K	40
systems	junior	21-25	45-50K	20
systems	senior	31-35	65-70K	5
systems	junior	26-30	45-50K	3
systems	senior	41-45	65-70K	3
marketing	senior	36-40	45-50K	10
marketing	junior	31-35	40-45K	4
secretary	senior	46-50	35-40K	4
secretary	junior	26-30	25-30K	6

Given a data sample with the values "systems", "junior", and "20-24" for the attributes department, status, and age, respectively, what would a naive Bayesian classification of the salary for the sample be?

Q-6

[A]

Compare the advantages and disadvantages of eager classification (e.g., decision tree, Bayesian, neural network) versus lazy classification (e.g., k-nearest neighbor, case-based reasoning). 4

[B]

A database has four transactions. min_sup = 60% & min_conf = 80%. 4

TID	Items bought
T1	{K,A,D,B}
T2	{D,A,C,E,B}
T3	{C,A,B,E}
T4	{B,A,D}

a. Find all frequent itemsets using Apriori algorithm.

OR

a. Find all frequent item sets using FP-growth with same database.

[C]

What are the similarity between classification and clustering? Can it be possible to use classification algorithms for clustering? Justify. 4

OR

[C]

Suppose the following table is derived by AOI. 4

CLASS	BIRTH PLACE	COUNT
	CANADA	180
Programmer	Others	120
DBA	CANADA	20
	Others	80

a. Transform the table into a cross-tab showing the associate t_weights and d_weights.